

Artificial Intelligence in Classrooms: Ethical Dimensions

KEY FINDINGS

- **Conceptual foundations:** To develop a robust ethical framework for AI in education depends on a robust understanding of AI ethics in general. In turn, this requires moving beyond commonsense notions to engage with formal normative frameworks, such as deontology and consequentialism.
- **Domain-specificity:** Education presents unique ethical challenges that general AI frameworks do not fully address, such as the specific purpose of education and the nature of useful knowledge. Because many users are legal minors, even standard ethical issues become acute.
- **The complexity of the AI pipeline:** Ethical risks are present at every stage of the AI pipeline, from the initial definition of tasks and data curation to model training and deployment. Issues such as the use of 'ghost workers' for data labelling need to be addressed from an ethical perspective.
- **Human agency:** The integration of AI in education must prioritise the professional autonomy of teachers and the independent development of students. Delegating pedagogical decisions to algorithms risks 'automated abdication' and 'cognitive atrophy'.
- **Equity and the flipped AI divide:** AI has the potential to exacerbate existing educational inequalities due to a gap in access to technology between different socio-economic groups, a gap that is beginning to flip so that students from higher-income backgrounds retain access to human teachers, while others are 'relegated' to AI-mediated instruction.
- **Child rights:** AI in education risks violating the *UN Convention on the Rights of the Child* (1989) by reducing students to data points, eroding autonomy, reinforcing biases, and facilitating commercial exploitation (treating children's interactions as unpaid labour to train proprietary models for profit).
- **AI literacy** is a foundational ethical requirement that should shift from teaching people how to use technology to helping them thrive and flourish in a world where AI is ubiquitous. Current AI literacy frameworks focus narrowly on the technological and practical dimensions of AI, omitting the human dimension and critical issues such as the impact of AI on democracy, social agency, economic inequalities, the environment, and geopolitical power struggles.
- **Governance:** To bridge the 'principles-to-practice' gap, enforceable governance mechanisms (such as mandatory auditing and risk assessments) are essential. A key aim is to shift the ethical burden from users to developers, while protecting teacher autonomy and student agency.



Introduction

For more than a decade, and accelerated by the arrival of generative AI (GenAI), a growing diversity of AI-mediated technologies have been widely deployed in classrooms and other education settings across Europe and beyond (Holmes et al., 2019; Tuomi, 2018). These include student-focused AI systems (from adaptive tutoring systems to support chatbots), teacher-focused AI systems (from plagiarism detection to resources curation), and institution-focused AI systems (from admissions to e-proctoring), together with other AI systems not designed specifically for education (from communication platforms to office software suites).

A European review (Holmes & Tuomi, 2022) identified twenty-five different types of AI-mediated technologies being used in education (AIED), many of which now have GenAI embedded, and all of which raise significant ethical questions. In fact, ethics permeates every aspect of AIED (from pedagogy to cognition, personalisation to data protection, and explainability to manipulation) and has implications for multiple stakeholders (including learners, teachers, schools, technology providers, educational authorities, EU Member States, and EU institutions).

The discussion of ethics and AIED is nothing new (Holmes et al., 2022; Holmes and Porayska Pomsta, 2023). Yet, the multiple AIED guidance documents published in recent years by international agencies, national agencies, NGOs, industry and academia rarely engage with ethics in any real depth. They all do mention ethics; however, they mostly adopt a 'commonsense' or intellectually naïve approach. They often appear unaware of two millennia of moral debate and the fundamental conceptual tensions; and they also almost always limit the ethics of AIED to the ethics of AI (usually in terms of bias, privacy and transparency), mostly ignoring both the ethics of education and the ethical questions that emerge from the interaction of AI and education.

The argument here is that for AIED ethical guidance to be actionable and effective, it is first necessary to go beyond the 'commonsense' superficial understandings of 'being ethical' or 'being good', to ground the discussion in a robust understanding of normative ethics – something that almost never happens. The fact that the many guidance documents tend to ignore two millennia of moral philosophy is a core weakness that inevitably compromises their implementation. Put another way, questions of implementation are irrelevant if what is to be implemented is poorly framed or understood. Second, for the ethics of AIED to be actionable and effective, it is also necessary to critically engage with – although not be limited to – the ethics of AI (addressing issues such as fairness and non-discrimination, data protection and privacy, autonomy and accountability, all of which are pertinent to all domains in which AI is applied). Third, for the ethics of AIED to be actionable and effective, it is also necessary to account for the ethics of education itself (issues such as the aims of education, the choice of pedagogy, teacher and student agency), issues that are too often ignored or unacknowledged.

For all these reasons, in order to examine how AIED may be developed, deployed and used ethically, this briefing document begins with the necessary foundations: a brief introduction to normative ethics, followed by an exploration of the ethics of AI, and finally a focus on the ethics of AIED itself. Not beginning with the specific ethical issues raised by AIED might seem counter intuitive; however, first clearly establishing the foundations (which are too often ignored) is essential.

Foundations

Normative Ethics

Any examination of the ethics of AIED should be grounded in core concepts from moral philosophy – which, given the long history of ethical debate, is not straightforward. Which ethical concepts from the history of moral philosophy should be prioritised, and how can they be used to disentangle and illuminate contemporary ethical questions in order to rationalise, defend, and recommend ‘good’ actions? The most cited normative ethics approaches in the European context are virtue ethics, deontology, and consequentialism. Each comes in multiple variants and has been debated for millennia.

Historically the first approach, virtue ethics focuses on moral characteristics or virtues (Aristotle, 4th Century BCE). Acting ethically means acting as a virtuous person would act, which requires determining what a virtuous person would do in any given situation. Example virtues include honesty, courage, compassion, generosity, fidelity, integrity, fairness, self-control, and prudence. The belief is that anyone who has developed these virtues will be disposed to act in ways that are ethical.

Deontology (D), which emerged two millennia after virtue ethics, sought instead to base morality on human reason rather than religious tradition or personal character (Kant, 1785). This approach emphasises universal human rights (such as those articulated in the Charter of Fundamental Rights of the European Union, 2000) and corresponding obligations. For example, every individual has both a right to respect and dignity and the obligation to afford respect and dignity to others. Accordingly, acting ethically means upholding obligations such as beneficence (doing good), non-maleficence (not causing harm), justice (ensuring people receive that which they deserve), and fidelity (honesty and keeping promises).

The third normative ethics approach is consequentialism (C), with utilitarianism being the best-known variant (Mill, 1863). For consequentialism, the outcomes or consequences of any action takes precedence over any rights or obligations; and acting ethically means undertaking actions that result in the ‘best overall result’, which is formalised in utilitarianism as being the ‘greatest good for the greatest number of people’. Actions are not judged by their adherence to fixed principles (or rights) but by their success in maximising collective ‘utility’ (benefit or happiness). As a consequence, this approach fails to protect individual rights, as it allows for the interests of a minority to be sacrificed for the benefit of the majority.

In contemporary ethical discourse, virtue ethics is often acknowledged as a foundational tradition but is rarely drawn on in practice (it is challenging to codify character or write a law that mandates a person to be wise). Instead, contemporary frameworks tend to rely on a pragmatic blending of deontology and consequentialism (D/C). This occurs as institutions attempt to protect universal human rights and duties, such as privacy and dignity, while simultaneously striving to achieve the ‘best overall result’ for the greatest number of people. Consequently, ethical guidelines often function as a hybrid system where fixed moral rules are used to set boundaries, but the specific actions within those boundaries are justified by their perceived social utility and outcomes. Nonetheless, setting boundaries and specifying actions remain challenging.

Finally, it should be noted that deontology and consequentialism represent a specific Judeo-Christian heritage that does not exhaust the global or even the European moral landscape. Contemporary Europe is a pluralistic tapestry of multiple faiths and secular convictions. Accordingly, an inclusive ethical approach must move beyond these traditional pillars to incorporate the insights of diverse religious and philosophical worldviews (e.g., Islamic Ethics, which includes a requirement

for absolute impartiality, even if a decision goes against oneself). Within the European tradition itself, notable alternatives have emerged that shift the focus from abstract rules to human relationships. The 'ethics of care', for example, emphasises the importance of relational ties, empathy, and the specific needs of others over impartial, universal principles (e.g., Gilligan, 1977). Acknowledging these varied perspectives is essential, to ensure that ethical debates are responsive to the complexity of a multicultural society.

Ethics of AI

Over the past decade, discourse centred on the ethics of AI has grown in prominence, beginning with researchers such as Boddington (Boddington, 2017) and institutions such as The AI Now Institute (2016). Around seven years ago, Jobin and colleagues (2019) identified 84 sets of principles of ethical AI, some complementary, some conflicting. These included the Asilomar AI Principles (2017), the Montreal Declaration of Responsible AI (2018), and the IEEE General Principles (2019). Each of these offer a sometimes muddled blend of deontological- and consequentialist-based principles. Around the same time, the European Commission's High-Level Expert Group on AI published the landmark *Ethics Guidelines for Trustworthy AI* (2018).

The *Ethics Guidelines for Trustworthy AI* (2018) offers guidance for developing and deploying AI systems based on rights as enshrined in the Charter of Fundamental Rights of the European Union (D). First, for AI to be considered trustworthy, the Guidelines state that it must satisfy three conditions throughout its entire 'life cycle' (from task specification through to system decommissioning). It must be lawful (comply with all applicable laws and regulations), robust (resilient from both a technical and social perspective to prevent unintended harms), and ethical (adhering to four ethical imperatives: respect for human autonomy (D), prevention of harm (C), fairness (D), and explicability).

The guidelines also offer a practical framework with seven mainly deontological requirements: AI systems should support fundamental rights and allow for human intervention; be technically robust and safe; ensure full respect for privacy and data protection; be transparent, focusing on traceability, explainability of decisions, and clarity about the AI system's capabilities; avoid unfair bias and ensure accessibility and universal design for all users; promote sustainability and consider the impact on society, democracy, and the environment; and include mechanisms for responsibility, auditability, and the reporting of negative impacts.

This approach is detailed here as a key example of the extensive emergent literature on the ethics of AI. However, it remains unclear how such principles might be actioned in specific AI designs and application contexts. A substantial part of this problem lies in the complexity of the moral frameworks themselves. A second part of the problem is the relativistic nature of the concepts involved, which tend to depend on socio-cultural norms, which are in themselves subject to change over time, and can be culturally dependent. For example, concepts of justice and fairness are continually evolving based on socio-political and subjective realities, conflicting interests and expectations, and the needs of different stakeholders.

Another part of the problem is the fact that AI is neither a single application (even in a single domain, there are multiple approaches and systems) nor a simple process. Every AI system, and every part of the AI pipeline, raises specific issues. A typical AI development process involves task definition; data curation (selecting, acquiring, pre-processing and labelling data); defining the AI model (selecting the approach, specifying its 'objective function'— its aim, and designing the algorithms); training and validating the model; deploying the model; and refining the model based on feedback (Cramer et al., 2019). Ethical questions emerge at each of these stages. For example, it is well known

that most GenAI systems were refined based on the involvement of poorly-paid workers in some global south countries, who were exposed to the systems' original uncensored and often deplorable outputs, so that the developers could put 'guardrails' in place to prevent the public from seeing similar outputs. Many of these 'ghost workers' of GenAI suffered mental health issues due to the materials they were monitoring; yet, without them, today's familiar GenAI systems would not be available to us. From a consequentialist perspective, there's little problem. Millions of GenAI users have 'benefited' while only a small number of ghost workers suffered. However, from a deontological lens, the human rights of the ghost workers have been fundamentally undermined.

Other ethical issues typically engaged with by the AI research community include bias and fairness. Bias in AI has long been the subject of research (e.g., Crawford, 2013), mostly from a deontological perspective, revealing many overlapping types of bias, each of which deserves specific attention. For example, there is 'historic' bias (such as cultural, and social stereotypes), 'measurement' bias (data inevitably oversimplifies what it represents), and 'deployment' bias (when the task that the model is designed to address is different from the task for which it is being used) (Bolukbasi et al., 2016). More recently, following the launch of GenAI, almost all of which was trained on the Internet together with all its vast array of inherent biases, there has been a renewed focus on solving bias technologically (e.g., Ferrara, 2023), although with little success.

Fairness, on the other hand, focuses on what and whose values are embedded in the technology, who is excluded as a result, and whether this is justified in the light of moral principles. Here, there is also a tension between the social- and computational-sciences, with social scientists typically rejecting the possibility of encoding fairness in an algorithm, given that fairness is a socio-culturally, institutionally, politically, and subjectively determined construct (Barocas et al., 2017). Two dimensions of fairness highlight the challenge. First, what may be considered fair for one group may be deemed unfair for another; second, should the focus be on outcomes (equity of results, which is a consequentialist lens) or process (equity of treatment, a deontological lens)?

Finally, having established this foundational overview of ethical questions emerging from the development and deployment of AI, it is also important to acknowledge that each domain in which AI is applied also raises specific ethical issues. For example, the ethics of AI in healthcare also raises ethical issues centred on, for example, patient choice; while the ethics of autonomous vehicles also involves ethical issues focused on assigning responsibility. Interestingly, this domain specificity is not substantively addressed by the EC's *Ethics Guidelines* (or by other guidance documents, such as Floridi & Cowlis, 2019). However, it is especially relevant to AI applied in education.

Ethics of AI in education

In education, pertinent although perhaps not immediately obvious ethical questions include the purpose and aims of education, what constitutes useful knowledge, the choice of pedagogy, the impact on human development, teacher and student agency, the lack of robust evidence for effectiveness or safety, and what it means to assess. All of these issues are relevant to the use of AI in education (in addition to the generic AI ethics issues), yet according to a recent systematic review of AI ethics in education, virtually none of these issues have been addressed in any depth (Yan et al., 2025).

Research into the ethical dimension of AI in education (AIED) has generally lagged behind that of other AI domains. Probably the first paper to examine the ethics of AIED in any depth appeared just a few years ago, *Ethics of AI in Education: Towards a Community-Wide Framework* (Holmes et al., 2021). Subsequently, however, there have been many related publications (e.g., Fu & Weng, 2024; Tang & Sheng Su, 2024; Williamson et al., 2023), including the first book *Ethics of AI in Education. Practices, Challenges and Debates* (Holmes & Porayska-Pomsta, 2023), along with guidance

documents from international agencies. One of the first of these guidance documents was UNESCO's *AI and Education: Guidance for Policy-makers* (2021), although it addressed ethics only in passing, which was followed by the EC's *Ethical Guidelines on the Use of AI and Data in Teaching and Learning for Educators* (2022) (with a new edition just released) and the OECD's *Opportunities, guidelines and guardrails for effective and equitable use of AI in education* (2023).

The EC's updated guidelines, *Ethical Guidelines on the Use of AI and Data in Teaching and Learning for Educators* (2026), offers a framework for the ethical and responsible integration of AI and data in educational settings, primarily targeting teachers and school leaders. The document covers three key areas: the core principles and legal context (including the EU AI Act and GDPR), guiding questions and practical scenarios for classroom application, and a suite of background resources and technical definitions. Key findings and recommendations emphasise that AI should remain human-centred (although what that means is never elaborated), serving as a tool to enhance rather than replace the teacher's role (which is not what most AIED tools do), while addressing generic AI risks such as algorithmic bias and data privacy, as well as the potential for over-reliance. The document also highlights five deontological ethical considerations (human dignity, fairness, trustworthiness, academic integrity, and justified choice) that it argues are essential for ensuring that AI technologies respect human rights and promote the best interests of all students (although human rights implications and the impact on disadvantaged students are rarely addressed by AIED developers). Unfortunately, however, the document also includes some unhelpful anthropomorphisms (such as using the inaccurate word 'hallucination' to mean the nonsense or confabulations that are frequently generated by GenAI). Anthropomorphising AI misleads users into over-estimating the capabilities and empathy of AI systems, and it could be argued has contributed to young people around the world committing suicide (Chatterjee, 2025) or dying by other means (e.g., drug overdoses, Black & Council, 2026).

Meanwhile, the OECD's *Opportunities, guidelines and guardrails for effective and equitable use of AI in education* (2023) focuses mainly on how to 'integrate' AI into education in order to leverage AI for 'personalised learning' and administrative 'efficiency'. In other words, it assumes that the integration of AI into education is ultimately effective (despite the lack of evidence to support such a view). It does mention the need for robust safety measures, to protect data privacy and ensure human agency, and it also stresses a 'human-in-the-loop' approach, asserting that AI must augment rather than replace the professional judgement of educators (which would actually require a 'human-in-command' approach) to maintain equity and ethical standards.

In summary, it is generally accepted by researchers, educators and policymakers that AIED raises far-reaching questions with important ethical implications for students, educators, parents, policymakers, and other stakeholders. The implications of the generic ethical issues mentioned earlier for AI (such as bias and fairness) are both relevant for education and even more acute, given that many users are legal minors (i.e., children in primary and secondary schools). However, as noted, there are also domain-specific questions, which so far have not been addressed in any depth (Yan et al., 2025).

These domain-specific questions include questions around the accuracy of diagnoses of learners interacting with AIED systems, choices of pedagogies employed by the systems, algorithmic predictions of learning outcomes, and questions related to the influence of AI on teachers' agency and decision making and on student autonomy and well-being. Interestingly, AIED also raises foundational questions such as: what is the purpose of education and hence the AIED tool? Is it to prepare students to pass examinations or to help them self-actualise? Is it to address issues identified in computer science departments or problems identified by educators in classrooms worldwide? Is it to replace teacher functions or to empower teachers? However, it still remains the

case that the ethical consequences of AIED for students, teachers and wider society are yet to be fully examined or detailed. All of these questions and more need to be addressed in any framework, if AIED is to be developed, deployed and used ethically.

Ethical risks of AIED throughout the AIED pipeline

In this section, a diversity of example ethical risks are identified (the list is far from comprehensive), arranged by their position in the AIED pipeline. These go beyond the typical focus of the generic issues for the ethics of AI (on, for example, bias, privacy and discrimination) to bring to the fore the specific ethical issues of AIED. The issues listed below, together with many others all need to be addressed in any effective ethical framework for AI in education (Holmes et al., 2025) – yet most of them are rarely mentioned (Yan et al., 2025). In particular, most are not addressed in any detail in either the EC's *Ethics Guidelines for Trustworthy AI* (2018) or their *Ethical Guidelines on the Use of AI and Data in Teaching and Learning for Educators* (2026). This warrants further examination (which will be undertaken in this project's follow-up study).

Example ethical issues emerging during the development of AIED

- **Representative learner datasets:** Ensuring models are trained on diverse student work, including non-native speakers and neurodivergent profiles.
- **IP of educational resources:** The ethical use of teacher-created lesson plans and academic journals to train models without fair attribution or compensation.
- **Data minimisation:** Avoiding requiring a student's intrusive personal data.
- **Stereotype threat:** Associating specific demographics with lower academic performance.
- **Environmental footprint:** The environmental cost of training AI models (due to the generation of large amounts of CO²) and running AI systems (which requires massive amounts of electricity and water), all of which conflicts with global citizenship and sustainability goals.

Example ethical issues emerging during the deployment of AIED

- **Pedagogical alignment:** The risk of system developers prioritising mathematical optimisation and broadly questionable pedagogies (e.g., repetition and memorisation) over well-established learning theories (e.g., constructivism, project-based learning, and productive struggle).
- **Algorithmic transparency:** Ensuring that educators are able to audit why a student was flagged as failing.
- **Child rights and data extraction:** Preventing corporate harvesting of student interactions for model development and commercialisation (i.e., preventing minors being commercially exploited).
- **Anthropomorphisation:** The systemic risk that, due to how AI systems are described and how AI outputs are presented, students are misled into attributing human consciousness, intentionality, and moral agency to AI systems.
- **The AI divide and flipped AI divide:** The gap in access to technology between learners from higher- and lower-socio-economic groups, which is morphing into a gap in access to human teachers: while students from higher socio-economic

groups continue to have access to teachers, students from lower socio-economic groups are increasingly having to make do with computers.

- **Accountability:** Automated abdication, where teachers rely solely on AI generated grades without exercising professional judgment.
- **Surveillance:** The use of AI to monitor eye-tracking, keystrokes or behaviour, leading to a climate of distrust and anxiety in the classroom.
- **Confabulations** (often referred to inaccurately and misleadingly as 'hallucinations'): The danger of AI systems presenting factually incorrect information or nonsense as 'truth'.
- **Impact on teacher workload:** The paradox where AI tools, meant to save time, actually increase the burden on teachers to verify, monitor, and troubleshoot automated outputs.
- **Lack of evidence:** The lack of independent evidence at scale for the effectiveness, safety or positive impact of AIED tools, meaning that most integrations of AI into education are effectively experimenting on children.

Example ethical issues emerging following the deployment of AIED

- **The deskilling of the profession:** The gradual loss of pedagogical expertise as teachers become facilitators of software rather than masters of their subject matter.
- **Cognitive dependency and atrophy:** The long-term erosion of foundational 'slow thinking' skills, such as critical analysis, as students outsource mental effort.
- **Erosion of social learning:** The loss of peer-to-peer collaboration and social-emotional development as education prioritises 'personalisation'.
- **Algorithmic determinism:** The danger of a student being locked into a specific trajectory based on an AI profile created during their early school years.
- **Homogenisation of student expression:** A narrowing of human creativity as students learn to write and think in ways that 'satisfy' the AI's statistical preferences.
- **Loss of cultural nuance:** The marginalisation of indigenous knowledge and minority languages.
- **Labour market misalignment:** The challenge that the education system might still be training students for roles that the AI they are using has already rendered obsolete.
- **The political economy of AIED and power dynamics in the technology ecosystem:** The concentration of AI infrastructure in a few global technology companies, and the growing dependence of education on proprietary AI platforms – i.e., the backdoor commercialisation of education, undermining its positioning as a common and public good (e.g., what happens when the commercial entity changes their pricing or withdraws products which have been embedded in education?).

Child rights

The use of AI in education also presents a profound challenge to the fundamental rights of children, as established by the UN Convention on the Rights of the Child (1989), necessitating a shift from speculative narratives of technological ‘potential’ toward a rigorous examination of socio-technical risks (Holmes, 2025).

At its core, the ethical use of AI depends on the protection of the human dignity and agency of children. However, with pedagogical decisions increasingly being delegated to algorithms, there is a significant risk that children are reduced to mere data points, undermined by systems that lack the human empathy and intuition essential for responding to their emotional and psychological needs. This algorithmic mediation of learning can further erode a child’s autonomy and their right to exercise individual choice. The deployment of AI systems in education also threatens a child’s right to be protected from discrimination, as biased data and historical stereotypes often perpetuate existing inequalities. And, finally, the invasive nature of AIED systems, which continuously monitor student behaviour and emotions, also creates a panoptic effect that compromises the right to privacy. Perhaps most critically, AIED systems also facilitate the commercial exploitation of children’s data. By extracting learner interactions to train proprietary models for profit, commercial developers in effect treat children as commodities and unpaid labour.

AI literacy

While at first glance AI literacy might seem tangential to the ethics of AI in education, it is in fact foundational – without a proper understanding of how AI systems function, individuals lack the agency to make informed ethical decisions.

A recent rapid review of AI literacy frameworks for the Royal Society (London) found broad agreement that AI literacy was important but that existing AI literacy frameworks were very limited (Hillman et al., 2025). In particular, in the 115 identified frameworks (which included, for example, ‘AILit’ from the EU and OECD, 2025), AI literacy was most often framed as being focused on teaching learners how to use AI, rather than helping them to thrive and flourish in a world in which AI is increasingly ubiquitous. The shift is important.

The review identifies three dimensions of AI: technological (how AI genuinely works), practical (how to use AI ethically, safely and responsibly) and human (the impact of AI on individuals, human rights, the environment, and society as a whole); and it found that almost all the identified frameworks focused mainly on the technological dimension, with some reference to the practical dimension, and only passing mentions of the human dimension, usually in terms of ethics instantiated only as bias and privacy.

Almost none of these AI literacy frameworks mentioned issues such as the impact of AI on democracy, governance, and the rule of law (e.g. AI in elections, concentration of power, digital sovereignty, global governance), on social agency and critical thinking (e.g. algorithmic nudging, filter bubbles, cognitive dependency, creative erosion, social isolation, cultural homogenisation), on economic inequalities and labour disruption (e.g., job displacement, gig economy, skill obsolescence, corporate dominance, global disparities, precarious work); on environmental and sustainability costs (e.g. carbon footprint, e-waste, resource extraction, greenwashing, water consumption, biodiversity risks); the cultural and linguistic impacts (e.g. cultural bias, historical distortion, digital colonialism); the psychological and behavioural effects (e.g. AI addiction, erosion of empathy, surveillance, decision fatigue, identity manipulation, trust decay, behavioural profiling, agency loss); geopolitics and AI power struggles (e.g. US-China rivalry, weaponised AI, AI as soft power, digital iron curtain, global south agency); or future scenarios (e.g. existential risks,

generational divides, new social contracts, spiritual impacts, dystopian safeguards, utopian potentials).

The assertion here is that whether young people and their teachers are aware of such issues is a profoundly ethical question. If they are only taught how to use AI, they are not able to make informed decisions about whether to use it or how to do so safely, responsibly, and ethically. Becoming literate in the ethics of AI and the other aspects of the human dimension of AI is not about putting the responsibility on the shoulders of the users. Instead, it is about empowering them to protect themselves, and is a prerequisite to thriving and flourishing in an age dominated by AI.

Governance

Having identified some of the many ethical issues raised in all stages of the AIED pipeline, it is necessary to consider practical governance mechanisms (Брітченко, 2025). Experience in operationalising ethics across various sectors clearly demonstrates that, without appropriate and well-designed mechanisms, ethical commitments risk remaining merely aspirational. To bridge the 'principles-to-practice' gap, comprehensive governance must incorporate robust policy tools throughout the AIED pipeline. This involves, for example, mandatory impact and risk assessments, ethics-based auditing, and the development of certification frameworks. Such mechanisms are essential for mitigating the risks embedded in AIED (from profiling to function creep), to help ensure that the entire AIED pipeline remains consistent with established ethical norms, and to shift the governance and ethical burden away from the user to the developers and vendors.

By integrating enforceable controls, educational institutions might protect the professional autonomy of teachers and the independent development of students, avoiding risks such as 'automated abdication' and 'cognitive atrophy'. Ultimately, such governance will help ensure that AIED moves beyond narrow technological dimensions to protect human rights, democracy, and social agency.

Conclusions

Developing an actionable and effective ethical framework for AI in education depends on a robust understanding of AI ethics in general, moving beyond commonsense notions to engage with formal normative frameworks such as deontology and consequentialism. It also requires a robust understanding of both generic AI ethical risks and education specific AI risks, and that ethical risks are present at every stage of the AI pipeline – from task definition and the use of 'ghost workers' to model deployment – while domain-specificity presents unique challenges regarding the purpose of education and the protection of legal minors.

It is clear that the use of AI must prioritise human agency, specifically the professional autonomy of teachers and the independent development of learners. Meanwhile, addressing equity requires mitigating the flipped AI divide, where students from higher-income backgrounds retain access to human teachers while others are relegated to AI-mediated instruction. AI literacy is also a foundational ethical requirement; without a comprehensive understanding of the human dimension and broader ethical implications (including the impact on democracy, social agency, and economic inequalities) students and teachers cannot make truly informed decisions about whether or how to use AI – mediated tools safely and responsibly. Finally, it is essential for robust governance mechanisms to be in place at every stage of the AIED pipeline.

In summary, for AIED ethical guidance to be actionable and effective, it is necessary to ground the discussion in a robust understanding of normative ethics, the ethics of AI, and the ethics of education

itself. This foundational work, particularly in terms of governance mechanisms at every state of the AI development pipeline, together with how ethics is operationalised in current EU guidelines and how ethics can be made genuinely actionable and constructive, will be examined in depth in this project's follow-up study.

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